

Exam Stat Mech 16-11-2023

November 2023

Summary of key formulas

$$\left(\sum_i x_i\right)^2 = \sum_i x_i^2 + \sum_{i \neq j} x_i x_j$$

$$\sigma(x) = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

$$c_v = \left(\frac{\partial \langle E \rangle}{\partial T}\right)_V$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{N,V}$$

$$\ln(N!) \approx N \ln N - N$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Problem: Random walker (2 points)

Consider a random walk in one dimension for which the walker at each step is equally likely to take a step with displacement anywhere in the interval $d - a \leq x \leq d + a$, where $a < d$. Each step is independent of others. After N steps the displacement of the walker is

$$S = X_1 + X_2 + \dots + X_N$$

where X_i is the displacement after the i -th step. After N steps,

- (a) What is the expected displacement in only one step?

- (b) what is the average displacement, $\langle S \rangle$, of the walker, and
(c) what is his standard deviation?

Solution: Random walker

- (a) Expected displacement in only one step $\langle X \rangle$

$$\langle X \rangle = \int_{d-a}^{d+a} xp(x)dx \quad (1)$$

where $p(x)$ is the probability density of the walker. Given that the steps are equally likely, this is uniform distribution that must be normalized in the interval $(d-a, d+a)$, i.e.

$$A \int_{d-a}^{d+a} 1dx = 1 \implies p(x) = \frac{1}{2a} \quad (2)$$

hence

$$\langle X \rangle = \frac{1}{2a} \int_{d-a}^{d+a} xdx = d \quad (3)$$

- (b) The average displacement, $\langle S \rangle$, of the walker Starting from the definition, and considering that we have a uniform distribution, we have

$$\begin{aligned} \langle S \rangle &= \int sp(s)ds = \int sds \int dx_1 p(x_1) \dots \int dx_N p(x_N) \delta(s - x_1 - \dots - x_N) \\ &= \int dx_1 p(x_1) \dots \int dx_N p(x_N) (x_1 + x_2 + \dots + x_N) = N \langle X_i \rangle = Nd. \end{aligned}$$

- (c) The standard deviation σ

In a similar way,

$$\langle S^2 \rangle = \int dx_1 p(x_1) \dots \int dx_N p(x_N) (x_1 + x_2 + \dots + x_N)^2.$$

We know that

$$\left(\sum_i x_i \right)^2 = \sum_i x_i^2 + \sum_{i \neq j} x_i x_j.$$

The number of pairs with non-equal i and j is $N(N-1)$. After averaging we obtain $\langle S^2 \rangle = N \langle X^2 \rangle + N(N-1) \langle X \rangle^2$. Then, to calculate $\langle X^2 \rangle$ let us specify the normalized probability as

$$p(x) = \begin{cases} \frac{1}{2a}, & \text{if } d-a \leq x \leq d+a \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$\langle X^2 \rangle = \frac{1}{2a} \int_{d-a}^{d+a} x^2 dx = d^2 + \frac{a^2}{3}.$$

Finally, summarizing the result:

$$\langle S^2 \rangle - \langle S \rangle^2 = N \left(d^2 + \frac{a^2}{3} \right) + N(N-1)d^2 - (Nd)^2 = \frac{Na^2}{3}$$

Then

$$\sigma = \sqrt{\langle S^2 \rangle - \langle S \rangle^2} = \sqrt{\frac{Na^2}{3}} = \frac{a}{\sqrt{3}} N^{1/2}$$

Problem: Paramagnet in the microcanonical ensemble (3 points)

Consider a paramagnetic solid, a system of N magnetic non-interacting atoms with a magnetic moment. The magnetic moment of each atom can point either $\uparrow$ or $\downarrow$. Each atom i can therefore be in two different states: $s_i = 1$ if it points $\uparrow$, and $s_i = -1$ if it points $\downarrow$. In presence of an external magnetic field h , the Hamiltonian of the paramagnet is $\mathcal{H} = -h \sum_i^N s_i$.

- Find the number of microstates compatible with a total energy E and its associated entropy for large N as a function of $n_\uparrow$, the fraction of atoms with magnetic moment pointing $\uparrow$.
- In magnetic systems, the observable that quantifies the alignment of the spins with the magnetic field is the average magnetic moment or magnetization $m = n_\uparrow - n_\downarrow$. Relate magnetization with the energy of the system and find the dependency of the magnetization with the temperature.
- Analyze the behavior of m in the limits $T \rightarrow \infty$ and $T \rightarrow 0$ and interpret these limits in terms of the number of spins pointing $\uparrow$ and $\downarrow$.

Solution:

- First realize that the energy $E = -h \sum_i s_i = -hN(n_\uparrow - n_\downarrow) = -hN(2n_\uparrow - N)$. The number of microstates compatible with energy E then the number of combinations of N spins in two groups $Nn_\uparrow$ and $N(1 - n_\uparrow)$ so that:

$$\Omega(E) = \binom{N}{Nn_\uparrow}$$

- Second, Using the definition of entropy in the microcanonical ensemble $S = k_B \log \Omega(E)$, we obtain that for the thermodynamic limit (N large and Stirling's approximation)

$$S = -k_B (n_\uparrow \log n_\uparrow + (1 - n_\uparrow) \log(1 - n_\uparrow)) \quad (4)$$

- Since $m = n_\uparrow - n_\downarrow$, we have that $E = -hN m$. Therefore to obtain $m(T)$ we have to use the relationship $(\frac{\partial S}{\partial E})_{V,N} = \frac{1}{T}$ to obtain that:

$$m(T) = \tanh \frac{h}{k_B T} \quad (5)$$

- Finally, if we take the limits $T \rightarrow 0$ and $T \rightarrow \infty$ we obtain that:

$$\begin{aligned} \lim_{T \rightarrow 0} m(T) &\rightarrow 1 & n_\uparrow &= 1, \text{ all spins are pointing } \uparrow \\ \lim_{T \rightarrow \infty} m(T) &\rightarrow 0 & n_\uparrow &= 1/2, \text{ half of the spins are pointing } \uparrow, \downarrow \end{aligned} \quad (6)$$

Problem: Canonical free energy (2 points)

Demonstrate that in the canonical ensemble $F = -k_B T \log Z$ starting from the Gibbs (or Shannon) definition of entropy.

Solution: Canonical free energy

$$\begin{aligned} S_{\text{shannon}} &= -k_B \sum_n P_n \log P_n = -k_B \sum_n \frac{\exp(-\beta E_n)}{Z} \log \left(\frac{\exp(-\beta E_n)}{Z} \right) \\ &= -k_B \sum_n \frac{\exp(-\beta E_n)}{Z} (-\beta E_n - \log Z) \\ &= k_B \beta \langle E \rangle + k_B \log Z \sum_n \frac{\exp(-\beta E_n)}{Z} \\ &= \frac{\langle E \rangle}{T} + k_B \log Z \end{aligned}$$

then $S_1 = \langle E \rangle / T + k_B \log Z$ or equivalently $-k_B T \log Z = \langle E \rangle - T S$ remind us the definition of Helmholtz free energy: $F = U - T S$. The free energy of the canonical ensemble is $F = -k_B T \log Z$.

Problem: Specific heat of solids (3 points)

Consider a solid with $N_A = 6.023 \times 10^{23}$ atoms/mol. Each atom vibrates about its equilibrium position with three position coordinates and three momentum coordinates. The appropriate coordinates after diagonalizing the Hamiltonian are referred to as “normal mode coordinates” in terms of which

$$E = \sum_{i=1}^{3N_A} \left(\frac{p_i^2}{2m} + \frac{1}{2} k_i q_i^2 \right) \equiv \text{Energy/mol} \quad (8)$$

knowing that The value of R at atm that is at standard atmospheric pressure is

$$R = 8.3144598 \frac{\text{Joule}}{\text{mol K}} \quad (9)$$

- Find the specific heat of the solid in the classical approximation, and its value at standard atmospheric pressure
- Find the specific heat of the solid in the quantum description, using the Einstein approximation for quantum vibrations: $k_i = k_0 = m\omega^2$
- Using the result of (b) compute the limiting cases of the high and low temperature T (for notation you can use $\frac{\hbar\omega}{k} = \Theta_E$ known as Einstein temperature)

Solution: Specific heat of solids

- using the equipartition theorem

$$\bar{E} = 3N_A \left(\frac{1}{2} kT + \frac{1}{2} kT \right) = 3RT$$

this leads to $c_v = \left(\frac{\partial \bar{E}}{\partial T} \right)_V = 3R \approx 25 \text{ Joules/mole K}$

- as we did for the single quantum oscillator

$$\bar{E} = 3N_A \hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right)$$

then

$$c_v = \left(\frac{\partial \bar{E}}{\partial T} \right)_V = \left(\frac{\partial \bar{E}}{\partial \beta} \right)_V \left(\frac{\partial \beta}{\partial T} \right)_V = -\frac{1}{kT^2} \left(\frac{\partial \bar{E}}{\partial \beta} \right)_V = -\frac{3N_A \hbar\omega}{kT^2} \left[\frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2} \right]$$

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$$c_v = 3R \left(\frac{\Theta_E}{T} \right)^2 \frac{e^{\Theta_E/T}}{(e^{\Theta_E/T} - 1)^2}$$

High T (Classical):

$$\frac{kT}{\hbar\omega} \gg 1 \Leftrightarrow \frac{\Theta_E}{T} \ll 1$$

$$c_v = 3R \left(\frac{\Theta_E}{T} \right)^2 \frac{(1 + \frac{\Theta_E}{T} + \dots)}{(\frac{\Theta_E}{T} + \dots)^2} \approx 3R$$

Low T (Not Classical):

$$\frac{kT}{\hbar\omega} \ll 1 \Leftrightarrow \frac{\Theta_E}{T} \gg 1$$

$$c_v = 3R \left(\frac{\Theta_E}{T} \right)^2 e^{-\Theta_E/T} \text{ (exponential decrease as } T \rightarrow 0)$$